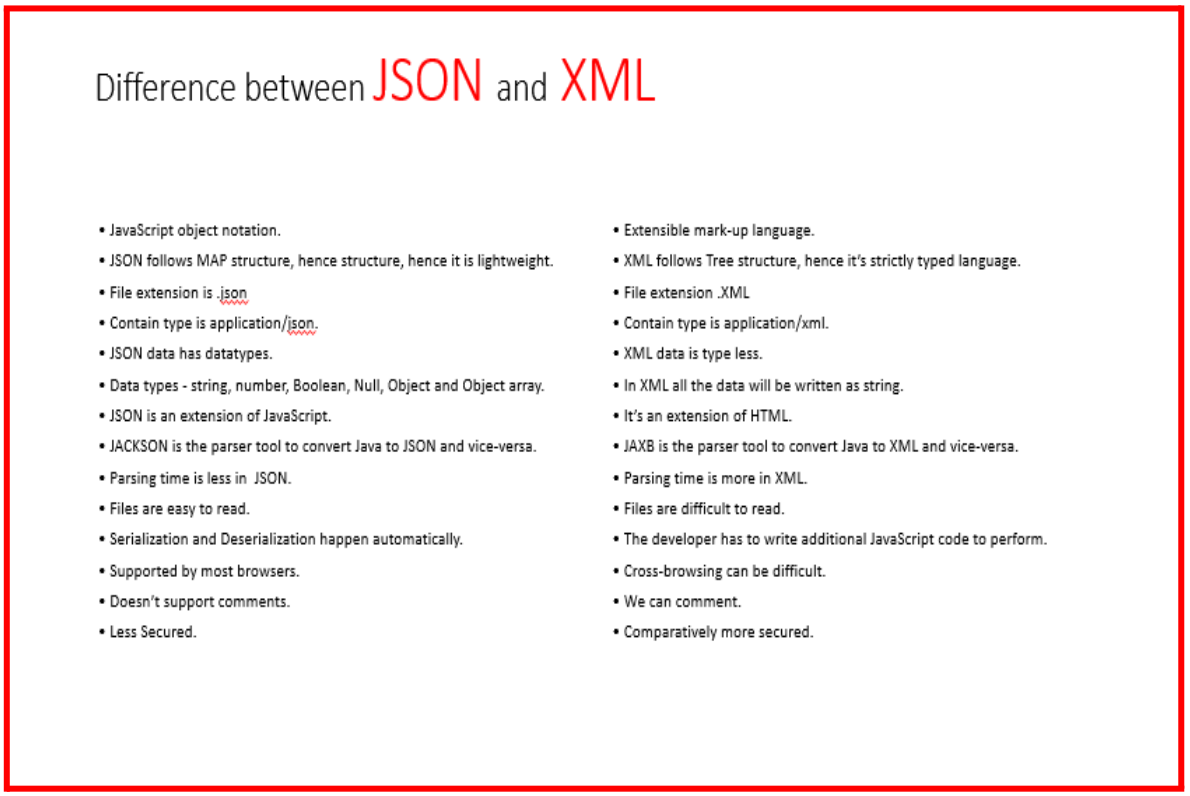
API NOTES



What is API?

Application Programming Interface acts like a bridge/ interface/ connection between two programs.

What is API Testing?

Testing the interface between two programs is called API testing.

Why API Testing is required?

API Testing is required in order to make sure that the Interface is working as expected with respect to Functionality, Performance, Scalability, Security etc.

Types of API Testing

- WBT (White Box Testing)
- Web Services Testing
  - SOAP Web Services Testing
  - REST Web Services Testing

White Box Testing

Testing each and every line of the source code using another program is called as WBT.

(OR)

Testing the untouched source code of the program is called as WBT.
WBT is done by Developer.

Web Services

- It's a mechanism where two applications or machines exchange information between each other irrespective of their underlined technology.
- So, web services help to share the functionality of one application with another application without sharing the source code or database.
- All the web services are exposed via API.
- Api are also known as web servives
- All api are not ws but ws are api

Who has to Test the API?

•The web service provider has to make sure that all the APIs are working as expected.

What is Web Services Testing?

•Testing the request and response between two applications is called Web Services Testing.

Why Web Services Testing is required?

•Web Services Testing is required in order to make sure that all the services are exposed by the provider are working as expected with respect to Functionality, Performance, Scalability, Security etc.

SOAP Web Service Testing

•SOAP is a protocol (Simple Object Access Protocol).

•It's an XML-based protocol used for applications to exchange information between each other.

•It will send the entire WSDL file (Web Services Description Language).

•UDDI: Universal Description Discovery Integration (It's where all the soap API's have to be registered).

What is SOAP Web Services?

•Two applications exchanging information between each other with the help of XML using SOAP as a protocol is called SOAP web service.

Note: All the SOAP Web Services are exposed via WSDL file.

What is SOAP Web Service Testing?

•Testing the request and response between two applications which is based on XML using soap as protocol is known as SOAP Web Service Testing.

What are the SOAP Web Service Testing tools?
•SoapUI
•SOAPSonar
•ReadyAPI
•Wizdler

REST Web Service Testing (Representational State Transfer):

•It's an architectural style used to develop Web Services.

•As the name suggests it represents the state of request.

•REST supports XML/HTML/JSON/Text/JavaScript.

•All the rest API's are exposed via URI (Universal Resource Identifier).

•URI=URL+URN (Universal Resource Name).

What is REST web service?

Two applications exchanging information with each other with the help of XML, HTML, Text, JSON, and JavaScript using HTTP as a protocol are called REST web services.

What is REST web services testing?

Testing the request and response between the two different application which is based on XML, JSON, JS, HTML, Text by using HTTP as a protocol is called REST web service testing.

What are the REST Web Service Testing Tools
•Postman
•FireFlink

•Rest Assured
•Ready API
•Karate

Difference between SOAP and REST Web Services

SOAP WEB SERVICE	REST WEB SERVICE
• It stands for Simple Object Access Protocol	• It stands for representation state transfer
• It's a protocol	• It's an architectural style
• It supports XML	• It supports XML/JSON/HTML/Text/JAVA SCRIPT.
• It uses SOAP as a protocol	• It uses HTTP as a protocol
• It's exposed via .wsdl file	• It's exposed via URI
• SOAP services are exposed as a package	• REST services are exposed as a get individual API's for each feature.
• It's designed using too many standards.	• It doesn't define too many standards.
• It defines its own security.	• It inherit to security measures like Basic Auth /Bearer Token/ OAuth1/OAuth2.0 /etc.
• SOAP is less preferred.	• REST is most preferred.

Advantages of Web Services:

1.Web Services Interoperability: •Web Services are "Application, Platform and Technology Independent".
•Ex: Uber/ OLA and Google Maps share the data.
2. Loosely Coupled : •Each application is independent of one another. Hence changes done to one application will Not Impact the "unrelated areas".

3.No need of re-inventing the wheel : •Web Services reduces the software development time.
•This helps the other business partners to quickly develop application and start doing business.
•This helps businesses To Save Time and money by cutting development time.
•Ex: Uber/ OLA can make use of Google Maps.

4.Business Opportunity : •Web Services will open the door for new Business Opportunities by making it easy to connect with partners.
•Ex: Dominos can get the order from Food Panda/ Swiggy along with getting orders from its own site.

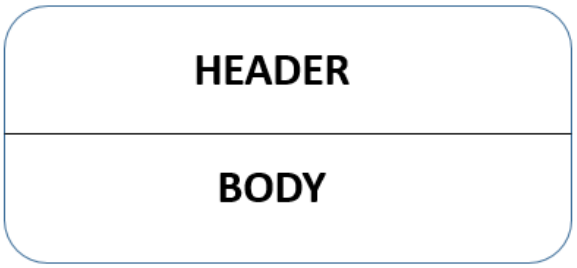
5.Service Reuse : •Web Services takes code Reuse a step further.

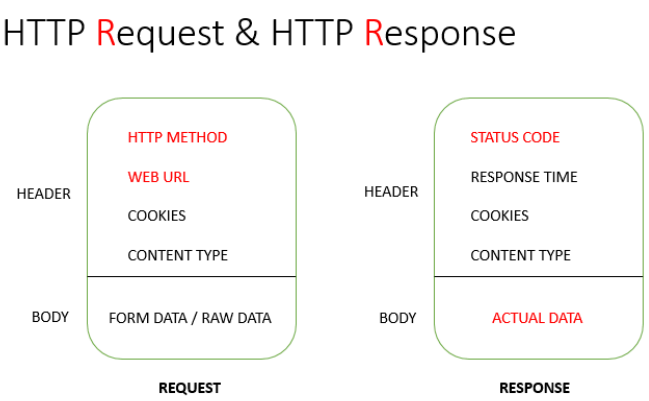
•Ex: An organization can have a “Single Payment Gateway Services” which helps other web applications of the organization to interact.

HTTP Structure:

•HTTP also has a structure and it consists of a header & body part separated by an empty line.

•In order to view the HTTP request & HTTP response structure we go for the Fiddler tool (developer debug tool).





HTTP Request - HTTP Method

	It's specifies the type of operation performed on the server.
HTTP Method	It's mandatory information present in the HTTP Request.
	HTTP method is the first element in the http request, used to specify the type of the request which is sent by client (Browser) to Server (Application).

HTTP Method

POST	It's used to create resources inside the server.
GET	It's used to read or retrieve the resources from the server.
PUT	It's used for a complete update of the resource inside the server and it can also be used for creating a resource.
PATCH	It's used for partial updating of the resources inside the server.
DELETE	It's used for deleting the resource inside the server.

Web URL:

•It's used to specify the location or address of the server.

•It's used to identify the specific web resource inside the web application.

What is Cookies:
It's an additional information provided.

Content Type

- It's used to specify the type of content used to transfer data.

Examples	application/json→.json
	Application/xml→.xml
	Application/zip→.zip
	Application/png→.png
	Application/html→.html

Form Data / Raw Data

- It's a data sent from the client to the server.
- GET will not have Form data.
- POST will have form data.

HTTP Response – Status Code

- It's used to specify the status of the request sent from the client.

100	200	300	400	500
• Continue.	• The server successfully handled the request.	• Re-direction request.	• Client-side error (requested resource not found on server side).	• Server-side error (Server encounter, an unexpected condition).

Response Time:

•The total time required to send the request, process the request, and get the response is called Response time.

What is Cookies:
It's an additional information provided.

Content Type

- It's used to specify the type of content used to transfer data.

Examples	application/json→.json
	Application/xml→.xml
	Application/zip→.zip
	Application/png→.png
	Application/html→.html

Actual Data:

•The data sent from the server to the client is called as Actual data.

Status Codes:

100 Series:
100 - Continue
101 - Switching protocols
102 - Processing
103 - Early Hints

200 Series:
200 - OK
201 - Created
202 - Accepted
203 - Non-authoritative info
204 - No Content
205 - Reset Content
206 - Partial Content

300 Series:
300 - Multiple Choices
301 - Moved Permanently
302 - Found
303 - See Other
304 - Not Modified
305 - Use Proxy

307 - Temporary Redirect
308 - Permanent Redirect

400 Series:
400 - Bad Request
401 - Unauthorized
402 - Payment Required
403 - Forbidden
404 - Not Found
405 - Method Not Allowed
406 - Not Acceptable
407 - Proxy Auth Required
408 - Request Timeout
409 - Conflict
410 - Gone
411 - Length Required
412 - Precondition Failed
413 - Payload Too Large
414 - URI Too Long
415 - Unsupported Media Type
416 - Range Not Satisfiable
417 - Expectation Failed
422 - Unprocessable Entity
426 - Upgrade Required
429 - Too Many Requests

500 Series:
500 - Internal Server Error
501 - Not Implemented
502 - Bad Gateway
503 - Service Unavailable
504 - Gateway Timeout
505 - HTTP Version Not Supported
507 - Insufficient Storage
511 - Network Authentication Required